

Research Proposal

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Preliminary title:

Coercion and Vulnerability: A District-Level Analysis of Rwanda's Crop Intensification Program (CIP) and Smallholder Resilience to Shocks.

Preliminary research question:

Does higher district-level land use consolidation intensity under Rwanda's Crop Intensification Program correlate with greater household food insecurity when households experience an adverse shock?

Subquestions

- Is the share of plots under land use consolidation (LUC) per district negatively associated with household dietary diversity scores among households that report experiencing an adverse shock in the past 12 months?
- Does the negative interaction between LUC intensity and shock exposure increase monotonically as household wealth decreases, indicating that the poorest households bear the greatest resilience cost?

Preliminary knowledge:

Puzzle and relevance. Rwanda's Crop Intensification Program (CIP), launched in 2007, is one of the most coercively enforced agricultural transformation programmes in sub-Saharan Africa. Under the programme's land use consolidation (LUC) component, farmers were mandated to cultivate state-assigned monocrops on consolidated plots, with compliance enforced through Rwanda's *imihigo* performance contract system—a cascade of accountability running from the president through ministers and mayors down to individual farmers (Cioffo et al., 2016). The programme produced nationally reported yield increases for priority crops and is widely cited as a demonstration of what strong state capacity can achieve in agricultural development (Cantore, 2011).

Yet this aggregate productivity narrative conceals a distributional question: *who paid the price for these gains?* Del Prete et al. (2019), using Rwanda's Integrated Household Living Conditions Survey (EICV), find that land consolidation increased production of roots and tubers while reducing dietary diversity—specifically decreasing consumption of meat, fish, fruits, and micronutrient-rich foods. Clay and Zimmerer (2020) go further, arguing through multi-season fieldwork that the CIP reduced smallholder *resilience* by stripping farmers of the crop diversity, livelihood flexibility, and resource access that previously buffered them against shocks. Their claim—that CIP households are more exposed to food insecurity when adversity strikes—has not been tested quantitatively. This study addresses that gap.

Theoretical approach. The core argument is that Rwanda's *imihigo* enforcement machinery enabled the state to impose a macro-level productivity goal by removing the micro-level adaptive strategies smallholders use to manage risk. The result is a structural vulnerability: CIP households are locked into state-assigned crops with reduced ability to diversify their diets or adjust

cultivation in response to adverse conditions—a cost that falls hardest on those with the fewest alternative resources.

Hypotheses.

- H1** Households in districts with higher LUC intensity will show lower dietary diversity scores when they experience an adverse shock than households in low-intensity districts (*resilience-reducing effect of CIP*).
- H2** The vulnerability-amplifying effect of LUC intensity will increase as household wealth decreases (*poverty-concentration hypothesis*).

Operationalization. The key independent variable is **district-level LUC intensity**, defined as the share of sampled agricultural plots recorded as located within an official land consolidation site. This is constructed from variable **s2q12** (“Is this plot located in a land consolidation site?”) in the NISR Seasonal Agricultural Survey (SAS) 2024 plot-level microdata, aggregated to the district level. The primary dependent variable is **household dietary diversity score (DDS)**, constructed from the food item consumption module of the EICV7 (2023/24) household survey (**CS_S8B_Food_Expenditure_Consumption.dta**), by collapsing individual food items into standard food groups and counting the number of groups consumed. The **shock variable** is a binary indicator derived from **s5eq1** (whether an unusual negative situation affected the household) and **s5eq2a–s5eq2c** (the nature of up to three severe shocks experienced in the past 12 months) in the EICV7 household module. The unit of analysis is the **household**, nested within districts ($N \approx 14,000$ households across 30 districts).

Preliminary structure:

1. Introduction
2. Literature review (focus on Clay & Zimmerer, 2020; Del Prete et al., 2019)
3. Theory and hypotheses: The Trade-off between Specialisation and Resilience
4. Empirics
 - 4.1 Research design and operationalization
 - 4.2 Methodology
 - 4.3 Results
5. Conclusion

Methodology note:

The study employs OLS regression with district fixed effects, estimated on the EICV7 (2023/24) cross-sectional household sample matched to district-level LUC intensity from the SAS 2024. The baseline model is:

$$\text{DDS}_{ihd} = \beta_1 \text{LUC}_d + \beta_2 \text{Shock}_{ih} + \beta_3 (\text{LUC}_d \times \text{Shock}_{ih}) + \mathbf{Z}'_{ih} \boldsymbol{\gamma} + \alpha_d + \varepsilon_{ihd} \quad (1)$$

where DDS_{ihd} is the dietary diversity score for household h in district d ; LUC_d is the district share of plots under land consolidation from SAS 2024; Shock_{ih} is a binary indicator equal to one if the household reported an adverse shock in the past 12 months; β_3 is the key coefficient (a negative value supports H1); $\mathbf{Z}_{ih}$ is a vector of household-level controls (household size, asset index, urban/rural residence); α_d are district fixed effects; and ε_{ihd} is the error term clustered at the district level. To test H2, the model is extended with a triple interaction term $\text{LUC}_d \times \text{Shock}_{ih} \times Q_{ih}$, where Q_{ih} is the household consumption quintile.

Data sources:

Variable	Source	Access
Dietary diversity score (DV)	EICV7 (2023/24), CS_S8B_Food_Expenditure_Consumption.dta	World Bank Microdata Catalog; free registration
Household shock indicator	EICV7 (2023/24), CS_S01_S5_S7_Household.dta, variables s5eq1–s5eq2c	World Bank Microdata Catalog; free registration
LUC intensity (% plots; s2q12)	NISR Seasonal Agricultural Survey 2024 microdata	NISR microdata catalog; free registration
Household controls	EICV7 poverty file (CS_EICV7_poverty_file.dta)	Included in EICV7 package
District merge key	Common to EICV7 and SAS 2024	Included in both datasets

Literature:

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